

TCS Senior .NET Full Stack Developer - What to Learn

A note before the keywords

This posting asks for 5-12 years for a "Senior" role – above a fresher/early-career profile. Still worth applying, but the seniority gap (not just skills) is the real hurdle here – no study guide fixes that.

Why these keywords

The stack pairs .NET/FastAPI backends with React/Angular frontends, PostgreSQL via ORMs (SQLAlchemy or EF Core), Playwright testing, and Azure DevOps – for a crypto-lending initiative. Your React and Python experience transfer directly; the delta is .NET, the two ORMs, Playwright, and Azure-specific tooling.

MUST learn (Class A – explicitly required)

.NET – Microsoft's back-end framework/runtime (C# is the language). Bridge: ASP.NET Core Web API controllers map closely to Express.js route handlers you already write. 20%: create a minimal ASP.NET Core Web API project with a GET/POST controller and DI.

FastAPI – a modern Python API framework using type hints for validation and auto OpenAPI docs. Bridge: a short hop from Flask – similar decorator routing, async-first, built-in Pydantic validation. 20%: build one FastAPI app with a Pydantic model and 2-3 endpoints.

SQLAlchemy & EF Core – Python's (SQLAlchemy) and .NET's (EF Core) ORMs, similar in spirit to Prisma. 20%: define one model class in each and run a basic query.

Playwright – browser automation/E2E testing (like Cypress/ Selenium, faster, multi-browser). A genuine gap (no tests in your repos today). 20%: write one test that navigates a page and asserts visibility.

Azure DevOps & App Services – CI/CD + project management (like GitHub Actions + Jira) and PaaS hosting (like Vercel). 20%: know Pipelines run YAML CI/CD jobs; App Services is where a built app deploys.

Veracode, OWASP, Key Vault – Veracode = security scanning (like SonarQube, security-focused); OWASP compliance = following the OWASP Top 10 (you already apply parts via parameterized queries, JWT); Key Vault = Azure's secrets service, direct analog of AWS Secrets Manager (already used in Echo). 20%: skim the OWASP Top 10 once.

Vite – fast frontend build tool/dev server, alternative to webpack. 20%: scaffold one app with `npm create vite@latest`.

GOOD to know / already strong (genuine strength)

GitHub Copilot / AI-assisted development – genuine daily user of Claude Code and Cursor AI; the JD explicitly wants developers "experienced with GitHub Copilot" – lead with this, workflow is identical.

React, Tailwind CSS, Node.js, PostgreSQL, Python, Git – already strong, no new prep needed.

Priority if time is short

1. FastAPI (fastest to pick up given your Flask/Django familiarity). 2. .NET Web API basics (biggest conceptual gap, but well-documented). 3. Playwright – one working test end to end. 4. Azure DevOps/App Services – just enough to discuss conceptually, by analogy to GitHub Actions + Vercel.